



Hamilton School for
Classical and Civic Education
UNIVERSITY *of* FLORIDA

Karl Marx and the Close of His System

*A 40-year controversy, 1867–1907: Böhm-Bawerk vs. Marx
(with Hilferding & Bortkiewicz) — ISS 2440, Intro to PPEL*

Louis Rouanet

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Marx, *Capital*, Vol. I

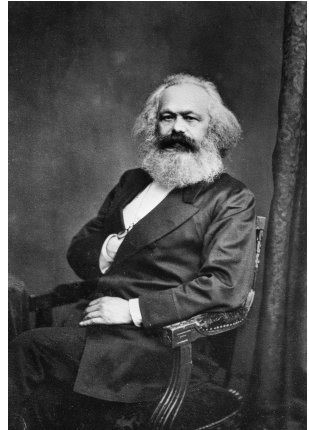
“Capital is dead labour which, vampire-like, lives only by sucking living labour, and lives the more, the more labour it sucks.”

His thesis in one line

Workers create **more** value than their wages are worth.
The capitalist keeps the difference.

Profit = **unpaid labour** = “surplus value.”

*The question this whole lecture answers: is profit **really** stolen labour?*



Karl Marx (1818–1883)

Two barrels of wine

A barrel of this year's **young** wine and a barrel of **aged** wine can hold the **same labour** — yet the aged one sells for far more. The extra value comes from **time**, not extra labour.

So where does profit come from?

The worker is paid **today**, but the product sells only **later**. The capitalist advances the wage and **waits**. Profit is the premium for that waiting — because people value goods **now** over the same goods later.

This is Böhm-Bawerk's own positive view, from his theory of capital and interest.

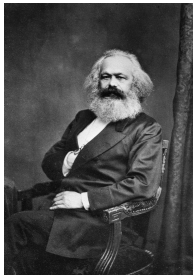


The clash

Marx: profit is **exploitation**.

Böhm-Bawerk: profit is the **price of time**.

Who is right?



Karl Marx
(1818–1883)
built the system



Böhm-Bawerk
(1851–1914)
the prosecutor



Hilferding
(1877–1941)
the defender



Bortkiewicz
(1868–1931)
the repairman

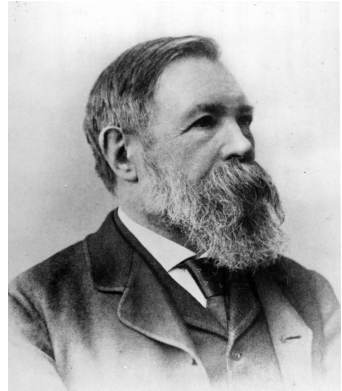
- ▶ Böhm-Bawerk: Austrian marginal-utility economist, three-time Austrian Finance Minister.
- ▶ Hilferding: Marxist, later twice German Finance Minister. Bortkiewicz: Russian-Polish statistician at Berlin.

The law of value

Vol. I (1867): commodities exchange in proportion to the **socially necessary labour** embodied in them.

- ▶ But Marx **admits** a clash with an everyday fact: capital of *any* make-up earns roughly the **same profit rate**, and goods don't sell at their labour-values.
- ▶ He calls the clash “only seeming” and **promises** to resolve it *later*.
- ▶ Marx dies in 1883 with the promise outstanding. Engels brings Vol. III to print in **1894**.

The whole controversy is a fight over **one IOU**.



Friedrich Engels (1820–1895)
published Vol. III in 1894

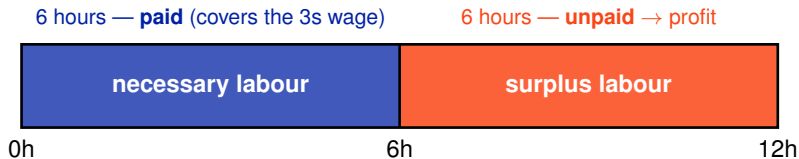
c — constant capital	v — variable capital	s — surplus value
machines & materials. Its value reappears unchanged .	wages / labour power. The only thing that adds <i>new</i> value, according to Marx.	unpaid labour the capitalist keeps .

The identity stack you will see all deck long

$$\begin{aligned}
 \text{Value} &= c + v + s & \text{exploitation} &= \frac{s}{v} & \text{profit rate} &= \frac{s}{c + v} \\
 \text{price of production} &= (c + v)(1 + r)
 \end{aligned}$$

*Plain words: only living labour (**v**) creates new value; machines (**c**) just pass theirs on.*

Marx's mechanism. A worker is paid 3s a day — a day's subsistence, which takes 6 hours of labour to produce. But the boss keeps him working a 12-hour day.



Marx's claim

The first 6 hours pay back the wage; the second 6 hours are worked **for free**. **Profit is the value of the unpaid hours.**

UF | The same story, in shillings

New value the worker creates is **proportional to the hours worked**. Over a 12-hour day he creates **6s** of new value — but is paid only **3s**.

	New value created	Wage paid (v)	Surplus (s)
6-hour day	3s	3s	0s
12-hour day	6s	3s	3s

The punchline

The wage stays **3s** either way (a day's subsistence). The longer day adds new value **beyond** what it costs.

$$\text{Rate of exploitation} = s/v = 3/3 = \mathbf{100\%}.$$

UF | Two ratios you must not confuse

The #1 first-timer error — pre-empt it before the ladder.

Rate of surplus value (exploitation)

$$= \frac{s}{v}$$

How the **new pie** is split between worker and boss.

$$\text{Rate of profit} = \frac{s}{c + v}$$

The boss's return on **everything** tied up — machines + materials + wages.

One worked figure, two very different answers

With $c = 410$, $v = 90$, $s = 90$:

$$\frac{s}{v} = \frac{90}{90} = \mathbf{100\%}$$

$$\frac{s}{c + v} = \frac{90}{500} = \mathbf{18\%}$$

*Adding machines (c) dilutes the profit rate without touching exploitation.
This dilution is the seed of the whole contradiction to come.*

Hold exploitation **fixed** at 100% ($s = v = 50$ throughout). Only pile on machinery (c).

Case	$c:v$	c	v	s	Total capital	Rate of profit
a (low: mostly wages)	0:1	0	50	50	50	100%
b (medium)	4:1	200	50	50	250	20%
c (high: lots of machinery)	9:1	450	50	50	500	10%

“Organic composition” just means **how much machinery per worker**: high = lots of machinery per worker; low = mostly wages, little machinery.

The takeaway — so prices **MUST** deviate from values

Same workers, same exploitation — yet the profit rate **falls** as machinery (c) swells. A pure labour theory predicts the **machine-heavy** line earns only 10% while the **labour-heavy** one earns 100%.

Competition won't allow that. That is the puzzle Vol. III must solve.

Engels edits and publishes Vol. III from Marx's drafts. The promised solution:

- ▶ Competition throws **all** surplus value into one **social pool** ...
- ▶ ... and hands each capital its **size-proportional** cut, so every industry earns the same **average profit rate** r .
- ▶ Goods sell not at values but at **prices of production**.

$$\text{price of production} = \underbrace{(c + v)}_{\text{cost price}} + \underbrace{r(c + v)}_{\text{average profit}} = (c + v)(1 + r)$$

Which assumption does Marx drop?

In Vol. I goods sold at their **labour values**. In Vol. III he admits they actually sell at **prices of production** (cost + the average profit). So he **drops the founding assumption** that price tracks labour — the very thing the labour theory was built to claim.

Marx's five industries ("spheres"). Each capital = 100; exploitation 100% (so $s = v$).

M1 — value = c used up + v + s

Sph.	c	v	s	Value	Profit rate
I	80	20	20	90	20%
II	70	30	30	111	30%
III	60	40	40	131	40%
IV	85	15	15	70	15%
V	95	5	5	20	5%

The problem

If goods sell at their **labour values**, the profit rate **scatters** from 5% to 40% — purely because some industries use more machinery than others.

The average

Across all five, the average make-up is $78c + 22v$, giving an average profit rate

$$r = \frac{110}{500} = \mathbf{22\%}.$$

Competition equalises profit rates. Each industry now earns the **same 22%**.

M2 — price = cost price +22

Sph.	Cost price	+22 profit	Price	Dev. (P-V)
I	70	22	92	+2
II	81	22	103	-8
III	91	22	113	-18
IV	55	22	77	+7
V	15	22	37	+17

Reading the two new columns

Cost price = the **c** used up + the wages (**v**) — what the capitalist actually lays out to make the goods.

+22: each industry's capital is 100, so the average 22% profit adds the **same** +22 to every cost price.

The deviations +2, -8, -18, +7, +17 **cancel to 0**. Both of Marx's totals still hold (used-up basis): total price = total value = 422; total profit = total **s** = 110.

Rows count only worn-out **c**. Marx's quoted aggregate 610 = all employed **c**(390)+**v**(110)+**s**(110) — do not add up the Value column.

UF | The deviations cancel — is the law saved?

Recall the five spheres sold at *prices of production* (M2). The price–value deviations were +2, −8, −18, +7, +17. Two things happen at once:

The deviations net to zero

$$\underbrace{(+2) + (+7) + (+17)}_{+26} + \underbrace{(-8) + (-18)}_{-26} = 0$$

Over-charges exactly offset under-charges.

And $5 \times 22 = \mathbf{110}$, the total of all surplus value.

Marx's two invariance claims

1. total price = total value = **422**

2. total profit = total surplus value = **110**

“In the same proportion in which one part of the commodities is sold above its value, another is sold under it.” — Marx

Cliff-hanger

The deviations cancel; the totals are conserved. **Is the law saved?** Böhm-Bawerk says *no* — and so, later, does Bortkiewicz.

His 1896 book is, in effect, one long review with a single question: did Volume III keep Volume I's promise?

His method (Sweezy's image)

"... like a problem in arithmetic — if you find an error in the first line, the whole calculation is worthless."

- ▶ So he tests the **value theory** first. Finding it broken, he declares the whole system worthless.
- ▶ The book became "almost the official answer of the economics profession."
- ▶ The German title means simply "**On the Conclusion of the Marxian System**" — a verdict, not an obituary.

Four chapters: I–II expound value & the transformation; III the contradiction; IV the error & its origin.



Eugen von Böhm-Bawerk (1851–1914)

Böhm-Bawerk's central charge: Volume III does not *reconcile* Volume I — it **flatly contradicts** it.

- ▶ **Vol. I:** goods exchange by labour, and labour *alone*.
- ▶ **Vol. III:** goods exchange by prices of production, deviating from labour-values **necessarily and permanently**.

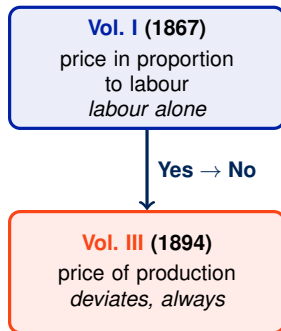
His either/or dilemma

Either products exchange in proportion to labour (then profit rates cannot equalize), *or* profit rates equalize (then products cannot exchange in proportion to labour). Both cannot hold.

Witnesses: Loria (“scientific suicide”); Sombart (“the knot was cut, not untied”).

Verdict

“Marx's third volume contradicts the first.” — Böhm-Bawerk



*They stand “as Yes to No —
as affirmation to contradiction.”*

Marx's first defense

Individual prices differ from values, but the differences cancel: **the totals still match** — total price = total value, and total profit = total surplus value. “So the law holds for the whole economy.”

Böhm-Bawerk's reply, in plain words

- ▶ A theory of value is supposed to explain **why any particular thing trades for what it does** — why this coat trades for that much linen.
- ▶ Saying “it all evens out across the whole economy” explains **no single real exchange**. People trade one good for another, not totals for totals.
- ▶ And in the table on the previous slide, **not one** of the five goods actually sells at its labour value.

Böhm-Bawerk's sharpest move against the "it holds on average" defense.

When does "on average" rescue a law?

Only if the ups and downs are **temporary wobbles** around a common center. But here the price–value gaps are **permanent and built-in** — not wobbles. "Marx confounds an average of fluctuations with an average between permanently and fundamentally unequal quantities."

The reductio

Elephants live ~ 100 years; mayflies ~ 1 day.

"Average" them to ~ 50 and declare all animals live equally long. **Averaging permanently unequal things explains nothing.**

A related jab

"Labour *affects* price" — everyone grants it.

That is **not** "labour *alone* determines price" — the actual claim.

Marx's third defense

Maybe goods exchanged at their labour values in **pre-capitalist** times, before competition equalized profit rates — so the law was once literally true.

Böhm-Bawerk: the history runs the other way

There is no evidence for it; it is just a hopeful re-description. And the historical record points the **opposite** way:

- ▶ Marx's theory predicts that in early capitalism the most **labour-heavy** industries should be the most profitable and grow fastest.
- ▶ But historically the **machine-heavy** industries — mining, chemicals, breweries — led the way instead.

Even after Marx’s rescue, the **size of profit** turns out to depend on things that are *not* labour at all.

Profit depends on more than labour

- ▶ the **rate of wages** (how much each worker is paid), and
- ▶ the **sheer amount of capital** advanced.

The point

Change the wage, or change how much capital is tied up, and profit moves — even with the labour input held fixed. So labour cannot be the **sole** determinant of profit.

The next slide shows the wage effect in a tiny example.

UF | Wages move prices on their own

Three goods, each priced at 100 — but with a **different make-up**: **A** a balanced mix, **B** machinery-heavy, **C** labour-heavy. Every price is just

$$\text{price} = \underbrace{\text{wages paid}}_{\text{rises with the wage}} + \underbrace{\text{profit}}_{= \text{avg. profit rate} \times \text{capital}}.$$

Now **raise the daily wage**, with the **labour input unchanged**. Two things happen at once: the wage part goes **up**, but the average profit rate **falls**, so the profit part goes **down**. Which wins depends on the good's make-up:

One up, one flat, one down

- ▶ **C** labour-heavy: wage rise wins → 100 → **108**
- ▶ **A** balanced: the two cancel → 100 → **100**
- ▶ **B** machinery-heavy: profit drop wins → 100 → **92**

Why it matters

If labour *alone* set prices, an unchanged labour input could not move a price at all. Yet the wage change moved all three. **Wages move prices on their own.**

Böhm-Bawerk's worked figures (wage 5 → 6, average profit rate 10% → 8%; full arithmetic in the report). Hilferding will dispute this example — we return to it.

Böhm-Bawerk moves from symptom to cause. There were three routes Marx *could* have taken to show that labour is the source of value — the question is which one he used.

Route 1 — Evidence

Compare real prices to labour. But Vol. III shows prices track total *cost*, not labour.

Route 2 — Motives

Examine why buyers & sellers trade. But motives push prices toward *usefulness* and *scarcity*.

Route 3 — Abstract logic

The one he uses: “exchange is an equation” $\Rightarrow$ a “common third thing” $\Rightarrow$ labour.

What “exchange is an equation” means — and why the step fails

When a coat trades for 20 yards of linen, Marx reads it as an equation: **coat = 20 yards of linen**. Just as “5 kg of apples = 5 kg of oranges” means both share a common measure (weight), he argues the two goods must share a common “third thing” — and declares it is **labour**.

Böhm-Bawerk: even if some common measure exists, nothing forces it to be **labour** — it could just as well be **usefulness** or **scarcity**. The conclusion does not follow.

Refutation stroke 1: the “common third thing” only *looks* like labour because Marx **rigged the field**. He admits only products of labour into the contest (“commodities”), silently excluding the **gifts of nature**:

- ▶ soil, trees, water power, coal beds, gold mines — bought and sold daily at non-arbitrary prices.
- ▶ Land’s price = capitalized rent, not “accidental.”



only *white* balls in

⇒ a white ball “must”
come out. So with labour.

Two images do the work

The urn: “he secures a white ball by putting in only white balls.”

The physicist: tests only *transparent* bodies, then announces transparency is the cause of *weight*.

Test a stacked sample, get a stacked answer.

UF | Is labour really the *only* thing in common?

Refutation stroke 2 — the line to memorize.

Marx's move

“Disregard use-value and there remains in commodities only **one** other property — that of being products of labour.”

Böhm-Bawerk: **Is there really only one?** Every exchangeable good is *also*

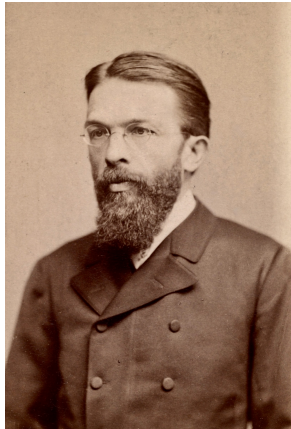
- ▶ scarce relative to demand; wanted (an object of supply and demand);
- ▶ owned (appropriated); a natural product; costly to produce.

Marx's only argument for labour is **negative** — but that same negative argument **kills labour too**.

Why this proves nothing

The very same kind of argument could just as easily crown **usefulness** as the common thing instead of labour. An argument that can prove either one proves *neither*.

Scarcity + usefulness is the door to the modern (marginalist) answer.



Carl Menger (1840–1921)

Skilled labor breaks the rule. A sculptor's day is worth far more than a stone-breaker's, with the *same* time worked. Marx's rescue: skilled labor “counts as” a multiple of unskilled labor.

To “count as” is not to “be”

A statistician cannot say that 100 deer plus 500 hares *contains* 1,000 hares.

“To ‘count as’ is not ‘to be,’ and the theory deals with the being of things.” — Böhm-Bawerk

The circle

Why does 1 sculptor-day count as exactly 5 stone-breaker days — not 3, not 10? Only because that is how they happen to trade.

“They exchange in this way . . . because . . . they do exchange in this way.” — the reasoning runs in a circle.

The takeaway

Marx *reads the multiplier off the prices* he is trying to explain — so it explains nothing.

Here is why the “counts as” move proves too much. Use Marx’s own reasoning, but put **quantity of material** in place of labor.

- ▶ 1 pound of raw gold trades for the same as a huge pile of iron — so gold “counts as” that much iron.
- ▶ By Marx’s logic we could then declare **material** the source of value — and call the figures at right a *confirmation*.

The point

The same trick “proves” material is the source of value just as well as it “proves” labor is. A test that confirms anything confirms nothing.

1 pound “counts as” ... of iron

1 lb raw **gold**
→ **40,000** lb iron

1 lb raw **silver**
→ **1,500** lb iron

1 **Cellini** gold cup
→ **4,000,000** lb iron

UF | Balanced forces still act

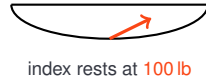
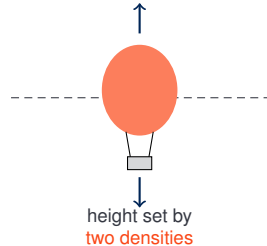
Marx belittles competition (supply & demand): when the two balance they “*cease to act*” and so explain nothing — leaving labor as the “real” cause.

Böhm-Bawerk: equilibrium is the *result* of the forces

Air balloon — it rises until its density equals the surrounding air's, and stops. Its height is explained **only** by the two densities.

Weighing scale — it rests at 100 lb when the weights balance. The reading is explained **by** the forces, not despite them.

- ▶ Same chapter, two opposite claims: competition pushes prices *toward* values *and away* toward prices of production.
- ▶ So where is the law of value ever in force? Nowhere.



Forces explain the equilibrium — they never stop acting.

UF | 1904: Hilferding strikes back

The only full-scale reply from the Marxist camp — and a real counter-punch. Hilferding says Böhm-Bawerk is **looking at value the wrong way**.

Two different starting points

Böhm-Bawerk starts from **one person's feelings** about how much a thing is worth.

Marx starts from **society as a whole**: in a market economy, labor is what links all the separate producers into one connected economy.

- ▶ So for Marx, value reflects a **social fact** — how a society's labor is shared out across all its products — not a private feeling.
- ▶ *"Not two methods, but two opposed and mutually exclusive outlooks upon the whole of social life."* — Hilferding



Rudolf Hilferding (1877–1941)

UF | Hilferding lands a real blow

The charge he answers: Böhm-Bawerk's wage example (slide 21) was meant to prove Marx's system **inconsistent** — wages moving prices in a way a labour theory can't absorb. **Hilferding's reply:** that only happened because Böhm-Bawerk *left the machinery cost (c) out*. Put it back — restock the inputs at their new prices too — and the books balance.

Then both totals hold

- ▶ total price = total value, *and*
- ▶ total profit = total surplus value.

And when wages rise

One good's price stays **flat**, one **rises**, one **falls** — and it all works through the new *average profit rate*, not as a separate “wage effect.”

Before the wage rise (c restored & fully used up)

	c	v	s	profit	Value	Price
A	450	50	50	50	550	550
B	670	30	30	70	730	770
C	230	70	70	30	370	330
Total	1,350	150	150	150	1,650	1,650

After the wage rise (5 → 6)

	c	v	s	profit	Value	Price
A	450	60	40	40	550	550
B	670	36	24	55	730	761
C	230	84	56	25	370	339
Total	1,350	180	120	120	1,650	1,650

One price **flat** (A), one **rises** (B), one **falls** (C) — yet the totals still match at 1,650.

What he answers

- ▶ With the machinery cost (c) restored, Marx's **books are consistent** after all.
- ▶ Value as a **social idea** — how society's labor is shared across its products — is coherent and well defined.
- ▶ Skilled labor can be treated as a multiple of ordinary labor.

What he leaves open

- ▶ He does **not** refute the point that the prices of different goods track **scarcity and usefulness**.
- ▶ He does not answer the **wine / time** example — equal labor, different prices, because of waiting.
- ▶ Whether *both* totals can hold at once when inputs too are priced consistently is left for **Bortkiewicz**.

UF | 1907: Bortkiewicz, the friendly repairman

A twist: a critic who wants to **save** Marx. Bortkiewicz — a Russian-Polish statistician — alone treated the surplus-value theory as Marx's *most important* contribution, not his Achilles' heel.

The diagnosis

Marx turned the **output** side into prices but left the **input** side (the **c** and **v** that firms buy) at the old *values*. Once goods sell at prices, inputs are bought at prices too — so they must be transformed as well. *“Like marking up your sale price for inflation but pretending your suppliers didn’t.”*

- ▶ Fix it on a 3-department scheme: I = means of production, II = wage goods, III = luxury / gold.
- ▶ The system has 3 equations but 4 unknowns, so you must add one **normalisation** — a choice of measuring unit. Bortkiewicz uses **gold** as the unit.



Ladislaus von Bortkiewicz
(1868–1931)

Three departments. Transform *both* sides — inputs as well as outputs. The consistent average profit rate comes out to $p = 25\%$.

At VALUES

Dept	c	v	s	Value
I (means of prod.)	225	90	60	375
II (wage goods)	100	120	80	300
III (luxury/gold)	50	90	60	200
Total	375	300	200	875

At CORRECT PRICES ($p = 25\%$)

Dept	c'	v'	profit	Price
I	288	96	96	480
II	128	128	64	320
III	64	96	40	200
Total	480	320	200	1,000

Now inputs are bought at prices too. Total profit = total surplus value = 200 **survives**, but total price = 1,000 $\neq$ total value = 875 — only **one** of the two equalities can be kept.

	Bortkiewicz (consistent)	Marx (Vol. III)
Average profit rate	25%	29.6%
Total profit = total surplus value?	YES (200)	YES (200)
Total price = total value?	NO ($1,000 \neq 875$)	YES (875) <i>but inputs untransformed</i>
Inputs (c, v) priced consistently?	YES	NO
Simple reproduction balances?	YES	NO

Lesson

Keep the books consistent and you can save **only one** of the two equalities:

1. total profit = total surplus value, *or*
2. total price = total value. **Not both.**

Bortkiewicz keeps (1) and lets total price drift from total value. What that means: the labour theory can still add up the economy's **total profit** — but it **cannot** tell you the price of one good against another (individual prices part ways from labour values). And Böhm-Bawerk was too harsh on one point: the transformation is *not* a hopeless muddle — Bortkiewicz showed it *can* be carried through consistently. The price of that consistency is giving up “total price = total value.”

Who won what

- ▶ **Böhm-Bawerk wins:** individual prices $\neq$ labor-values; labor is not the only thing all goods share; the wine / time counterexample stands.
- ▶ **Hilferding wins:** the system is internally consistent once the machinery cost (**c**) is restored, and value as a *social* category is coherent.
- ▶ **Bortkiewicz settles it:** the transformation can be done rigorously — but value magnitudes no longer add up to price magnitudes.

Net: the labor theory survives as **bookkeeping for surplus value**, not as a theory of **relative prices**.

Three discussion questions

- (1) Is the labor theory dead as a theory of prices but alive as a theory of **power**?
- (2) Does “the totals cancel” rescue a law that fails for every **individual** trade?
- (3) Why couldn't the two sides settle it with **evidence**?

Primary texts

- ▶ Böhm-Bawerk, *Karl Marx and the Close of His System* (1896).
- ▶ Hilferding, “Böhm-Bawerk’s Criticism of Marx” (1904).
- ▶ Bortkiewicz, “On the Correction of Marx’s Fundamental Theoretical Construction. . .” (1907).
- ▶ Marx, *Capital*, Volumes I & III.
- ▶ Böhm-Bawerk, *Positive Theory of Capital* (1889).